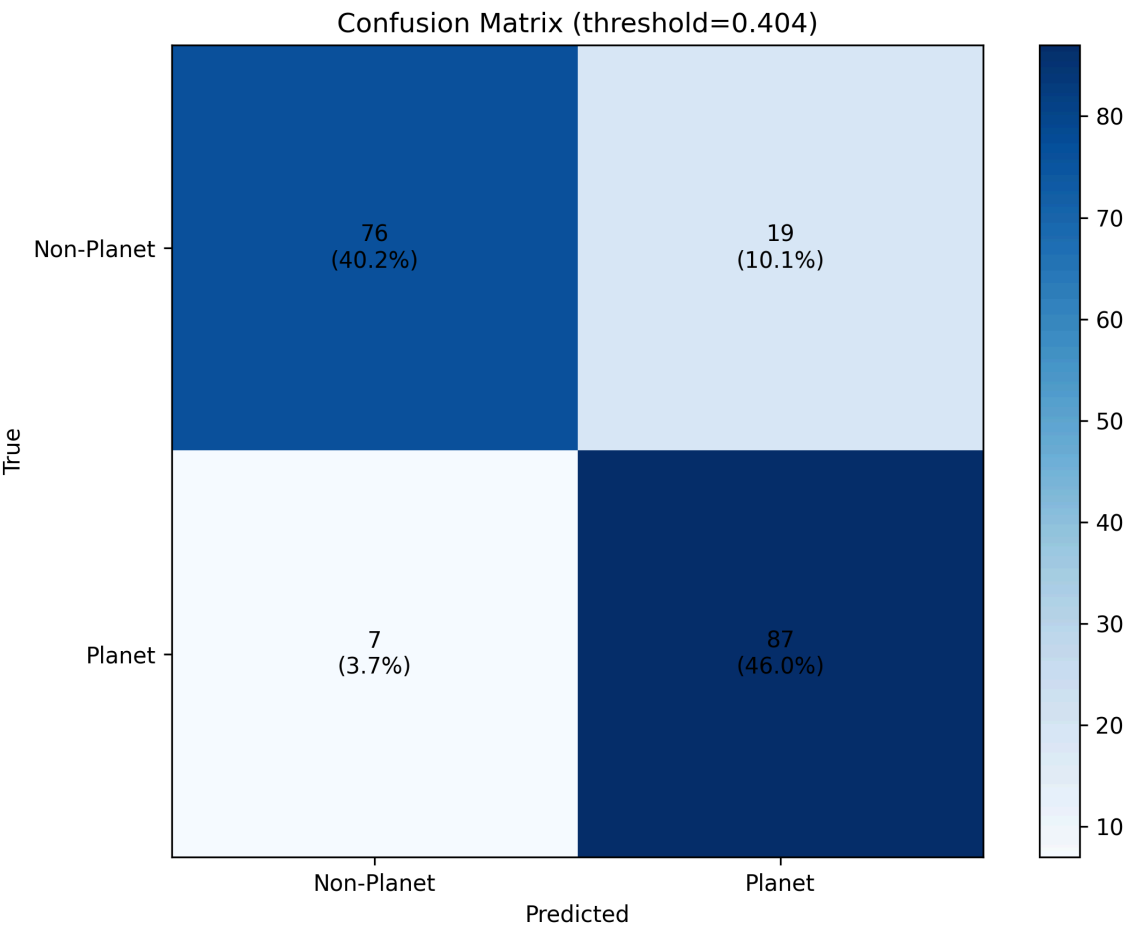


1) STANDARD RUN OF THE NOTEBOOK



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THRESHOLD OPTIMIZATION & EVALUATION

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Optimal threshold: 0.4042 (default=0.5)

At this threshold: TPR=0.9255, FPR=0.2000

Results:

AUC-ROC: 0.8979

Accuracy (default threshold=0.5): 0.7354 (73.54%)

Accuracy (optimal threshold=0.4042): 0.8624 (86.24%)

With optimal threshold:

	precision	recall	f1-score	support
Non-Planet	0.9157	0.8000	0.8539	95
Planet	0.8208	0.9255	0.8700	94

accuracy		0.8624		189
macro avg	0.8682	0.8628	0.8620	189
weighted avg	0.8685	0.8624	0.8619	189

Prediction distribution (optimal threshold):

Predicted 0: 83

Predicted 1: 106

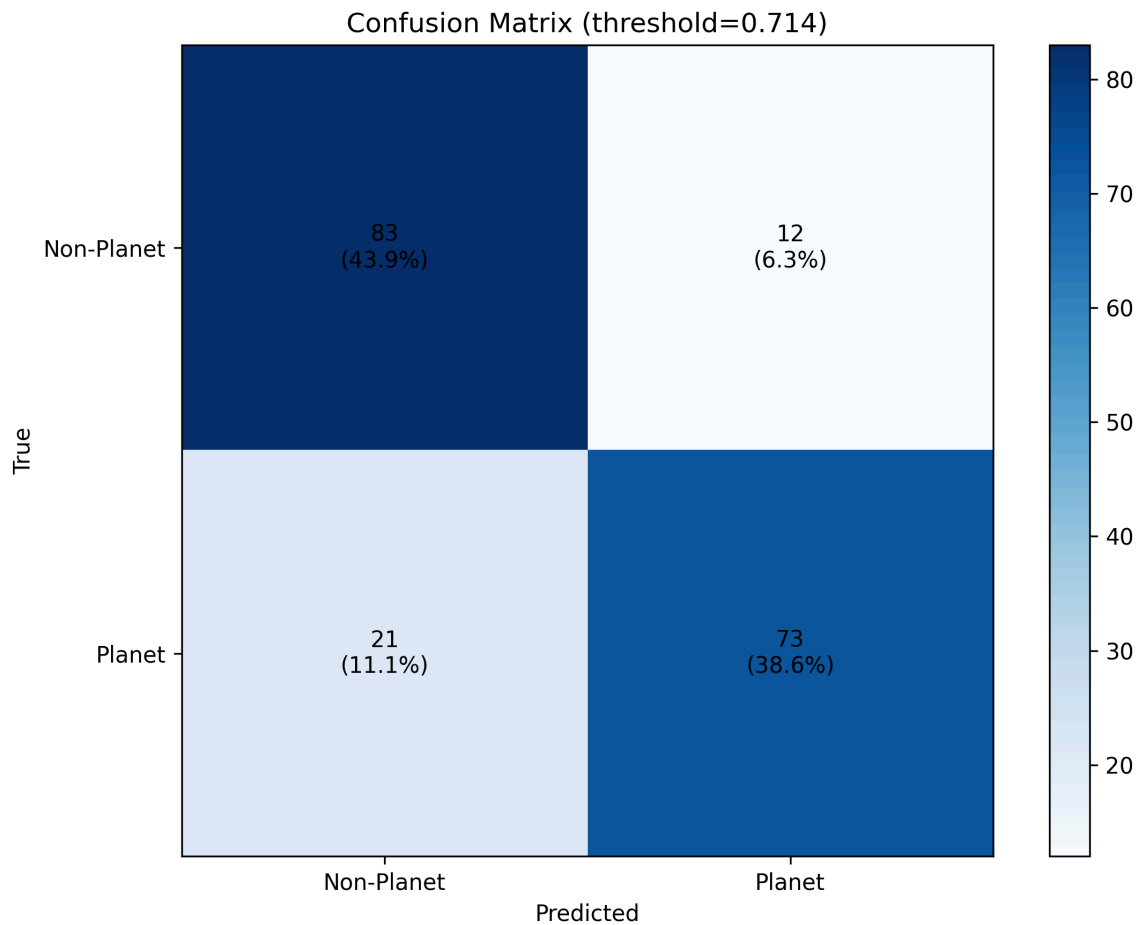
True distribution:

True 0: 95

True 1: 94

2) RUN WITH FOCAL LOSS PARAMETERS CHANGED (starting from default configuration)

$\alpha = 4$, $\gamma = 0.95$ instead of std ones



THRESHOLD OPTIMIZATION & EVALUATION

Optimal threshold: 0.7137 (default=0.5)

At this threshold: TPR=0.7766, FPR=0.1263

Results:

AUC-ROC: 0.8394

Accuracy (default threshold=0.5): 0.4974 (49.74%)

Accuracy (optimal threshold=0.7137): 0.8254 (82.54%)

With optimal threshold:

	precision	recall	f1-score	support
--	-----------	--------	----------	---------

Non-Planet	0.7981	0.8737	0.8342	95
------------	--------	--------	--------	----

Planet	0.8588	0.7766	0.8156	94
--------	--------	--------	--------	----

accuracy		0.8254		189
macro avg	0.8285	0.8251	0.8249	189
weighted avg	0.8283	0.8254	0.8250	189

Prediction distribution (optimal threshold):

Predicted 0: 104

Predicted 1: 85

True distribution:

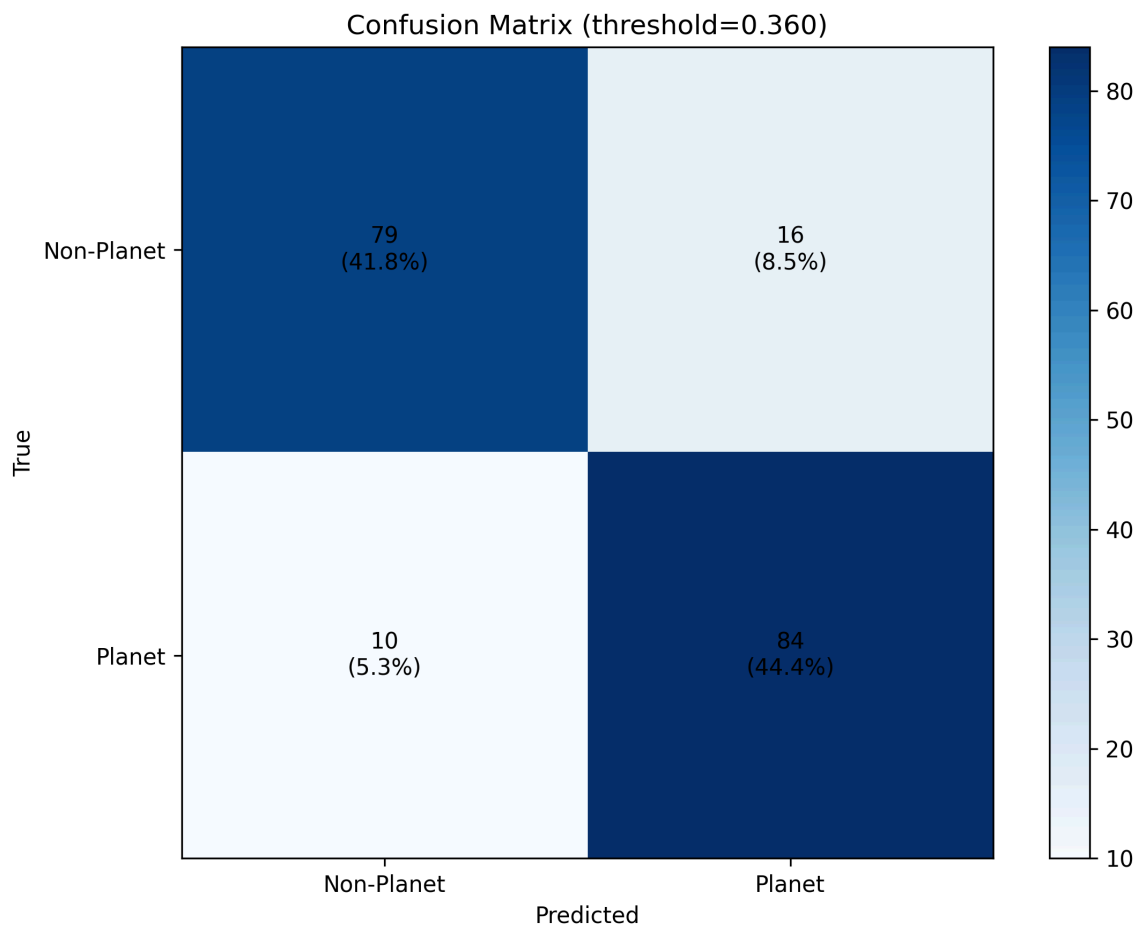
True 0: 95

True 1: 94

3) RUN WITH A NEW LAYER (Starting from default configuration)

Added code:

```
layers.Conv1D(512, kernel_size=3, padding='same', activation='relu'),  
    layers.BatchNormalization(),  
    layers.MaxPooling1D(2),  
    layers.Dropout(0.3),
```



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THRESHOLD OPTIMIZATION & EVALUATION

=====

Optimal threshold: 0.3603 (default=0.5)

At this threshold: TPR=0.8936, FPR=0.1684

Results:

AUC-ROC: 0.8772

Accuracy (default threshold=0.5): 0.6138 (61.38%)

Accuracy (optimal threshold=0.3603): 0.8624 (86.24%)

With optimal threshold:

	precision	recall	f1-score	support
Non-Planet	0.8876	0.8316	0.8587	95
Planet	0.8400	0.8936	0.8660	94
accuracy		0.8624		189
macro avg	0.8638	0.8626	0.8623	189
weighted avg	0.8639	0.8624	0.8623	189

Prediction distribution (optimal threshold):

Predicted 0: 89

Predicted 1: 100

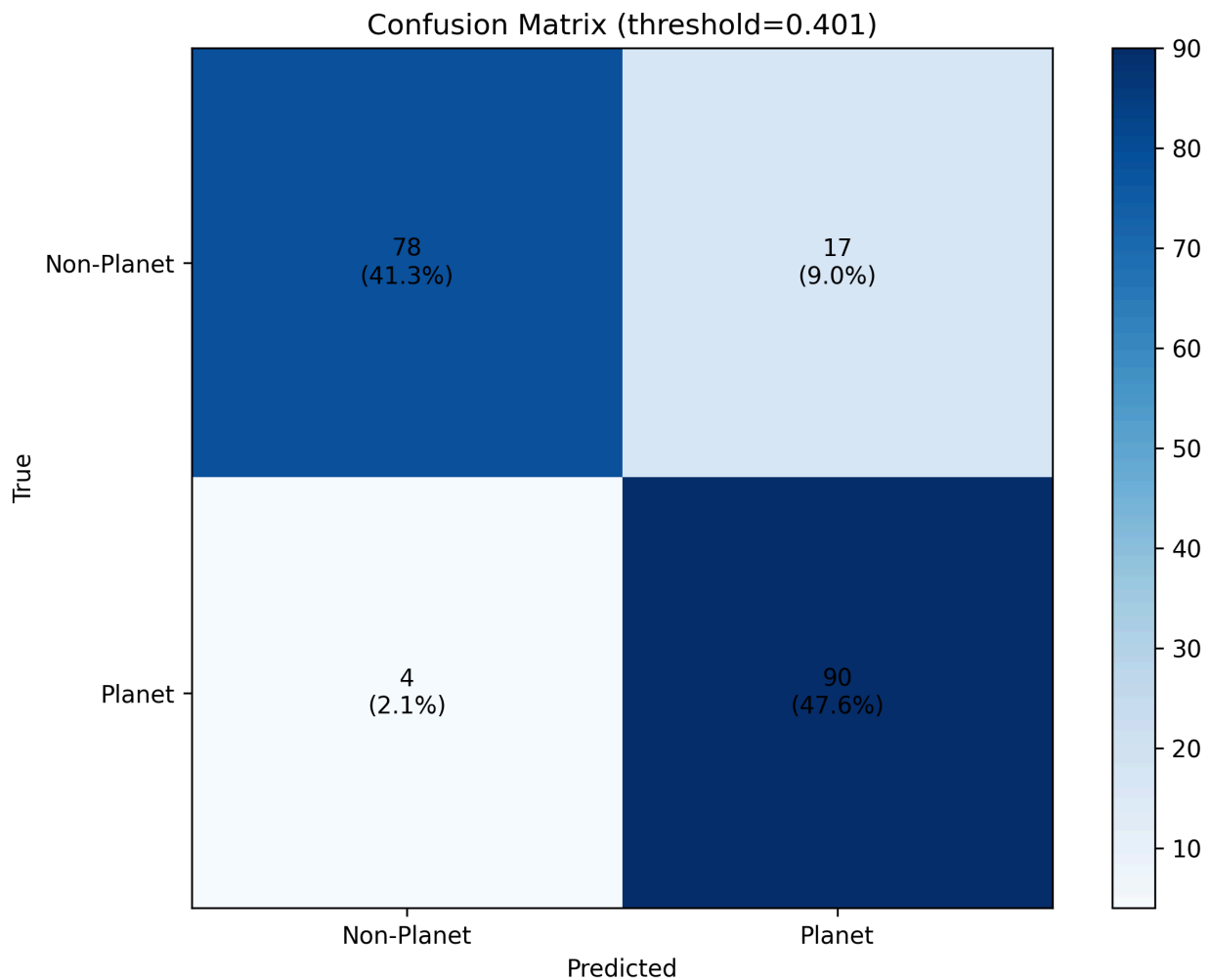
True distribution:

True 0: 95

True 1: 94

4) RUN WITH DROPOUT CHANGES AND LEARNING RATE CHANGE (Starting from Configuration 3)

Dropout systematically 0.1 smaller while 0.1 bigger for dense, learning rate from 0.0005 to 0.0002



=====

THRESHOLD OPTIMIZATION & EVALUATION

=====

Optimal threshold: 0.4006 (default=0.5)

At this threshold: TPR=0.9574, FPR=0.1789

Results:

AUC-ROC: 0.9151

Accuracy (default threshold=0.5): 0.7249 (72.49%)

Accuracy (optimal threshold=0.4006): 0.8889 (88.89%)

With optimal threshold:

	precision	recall	f1-score	support
--	-----------	--------	----------	---------

Non-Planet	0.9512	0.8211	0.8814	95
------------	--------	--------	--------	----

Planet	0.8411	0.9574	0.8955	94
--------	--------	--------	--------	----

accuracy		0.8889	189
----------	--	--------	-----

macro avg	0.8962	0.8892	0.8884	189
-----------	--------	--------	--------	-----

weighted avg	0.8965	0.8889	0.8884	189
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Prediction distribution (optimal threshold):

Predicted 0: 82

Predicted 1: 107

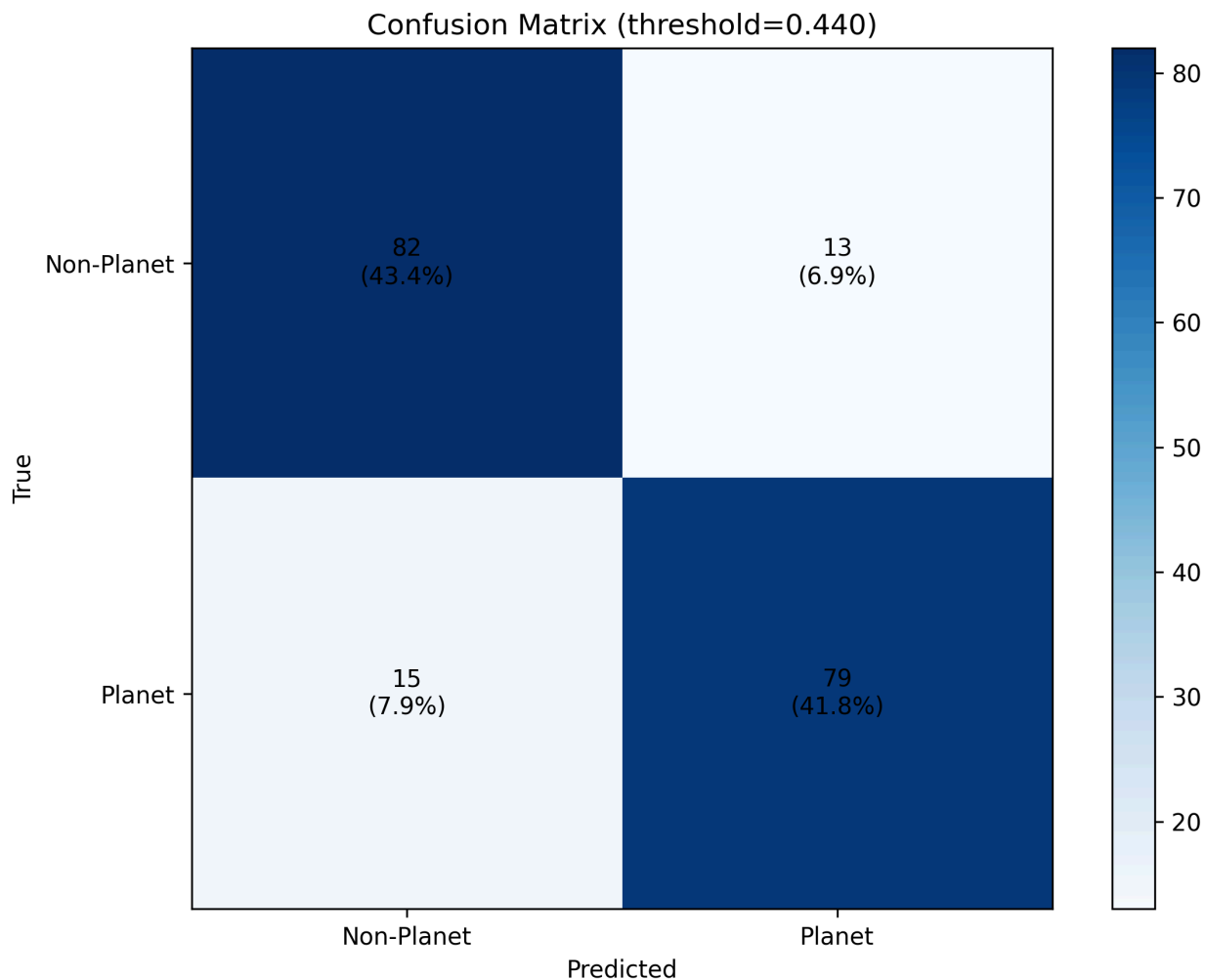
True distribution:

True 0: 95

True 1: 94

5) RUN WITH DROPOUT CHANGES AND LEARNING RATE CHANGE (Starting from Configuration 3)

Dropout systematically 0.1 bigger while 0.2 fixed for dense, learning rate from 0.0005 to 0.0003



THRESHOLD OPTIMIZATION & EVALUATION

Optimal threshold: 0.4405 (default=0.5)

At this threshold: TPR=0.8404, FPR=0.1368

Results:

AUC-ROC: 0.8997

Accuracy (default threshold=0.5): 0.7725 (77.25%)

Accuracy (optimal threshold=0.4405): 0.8519 (85.19%)

With optimal threshold:

precision recall f1-score support

Non-Planet 0.8454 0.8632 0.8542 95

Planet 0.8587 0.8404 0.8495 94

accuracy 0.8519 189

macro avg 0.8520 0.8518 0.8518 189

weighted avg 0.8520 0.8519 0.8518 189

Prediction distribution (optimal threshold):

Predicted 0: 97

Predicted 1: 92

True distribution:

True 0: 95

True 1: 94